KNIME PRESENTATION ON SALES DATA

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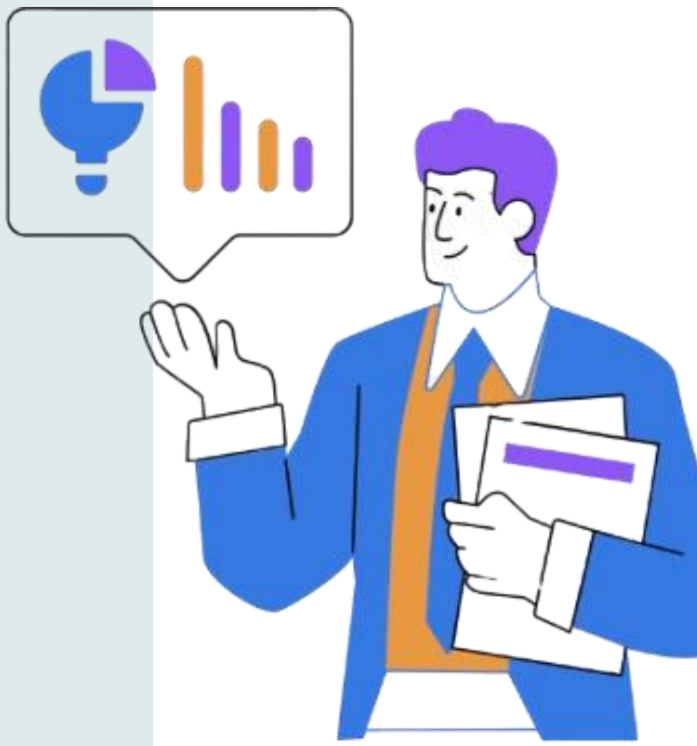


Part A

- **Business Context:**

- An automobile parts manufacturing company has been actively selling products to a diverse range of customers for the past three years. Despite its growth, the company lacks the in-house expertise to derive actionable insights from its transaction data. As a result, they wish to uncover hidden patterns and trends in their customer transactions. By analyzing this data, the company aims to better understand customer behavior, improve customer segmentation, and implement targeted marketing strategies. These insights will help the company not only enhance customer satisfaction but also drive revenue growth by offering more personalized and efficient services.

- **Objective:**

- The primary objective of this analysis is to leverage data science techniques to:

Identify underlying patterns in customer purchasing behavior. Segment customers based on their transactional data. Provide actionable insights to optimize the company's marketing efforts. Recommend personalized marketing strategies for each customer segment to maximize sales and customer retention. Your role as a Business Analyst is to use the provided dataset to achieve these goals and present findings in a manner that can guide the company's decision-making.

- **Data Description:**

- The dataset provided contains three years of transactional data from the company, with each row representing a unique order. Below is an explanation of the key attributes:

Columns	Description
ORDERNUMBER	Unique identifier for each order.
QUANTITYORDERED	Number of items ordered in a specific transaction.
PRICEEACH	Price per unit of the product in the order.
ORDERLINENUMBER	Sequence number of the product in the order.
SALES	Total sales value for the order.
ORDERDATE	Date when the order was placed.
DAYS_SINCE_LASTORDER	Number of days since the customer's previous order.
STATUS	Current status of the order (e.g., Shipped, Disputed).
PRODUCTLINE	Product category to which the item belongs (e.g., Motorcycles, Classic Cars).
MSRP	Manufacturer's Suggested Retail Price for the product.
PRODUCTCODE	Unique identifier for the product.
CUSTOMERNAME	Name of the customer placing the order.
PHONE	Customer's contact phone number.
ADDRESSLINE1	Customer's primary address.
CITY	City of the customer's address.
POSTALCODE	Postal code of the customer's address.
COUNTRY	Country of the customer's address.
CONTACTLASTNAME	Last name of the customer's contact person.
CONTACTFIRSTNAME	First name of the customer's contact person.
DEALSIZE	Size category of the transaction (e.g., Small, Medium, Large).

EXPLORATORY DATA ANALYSIS

The Classic car was the best-selling product, with a total of 33,373 units sold over the years. Vintage cars ranked second, with 20,059 units sold. Motorcycles, Planes, Trucks, and Buses each had sales ranging between 10,000 and 11,000. The least popular product lines were Ships, with 7,989 units sold, and Trains, with just 2,712 units.

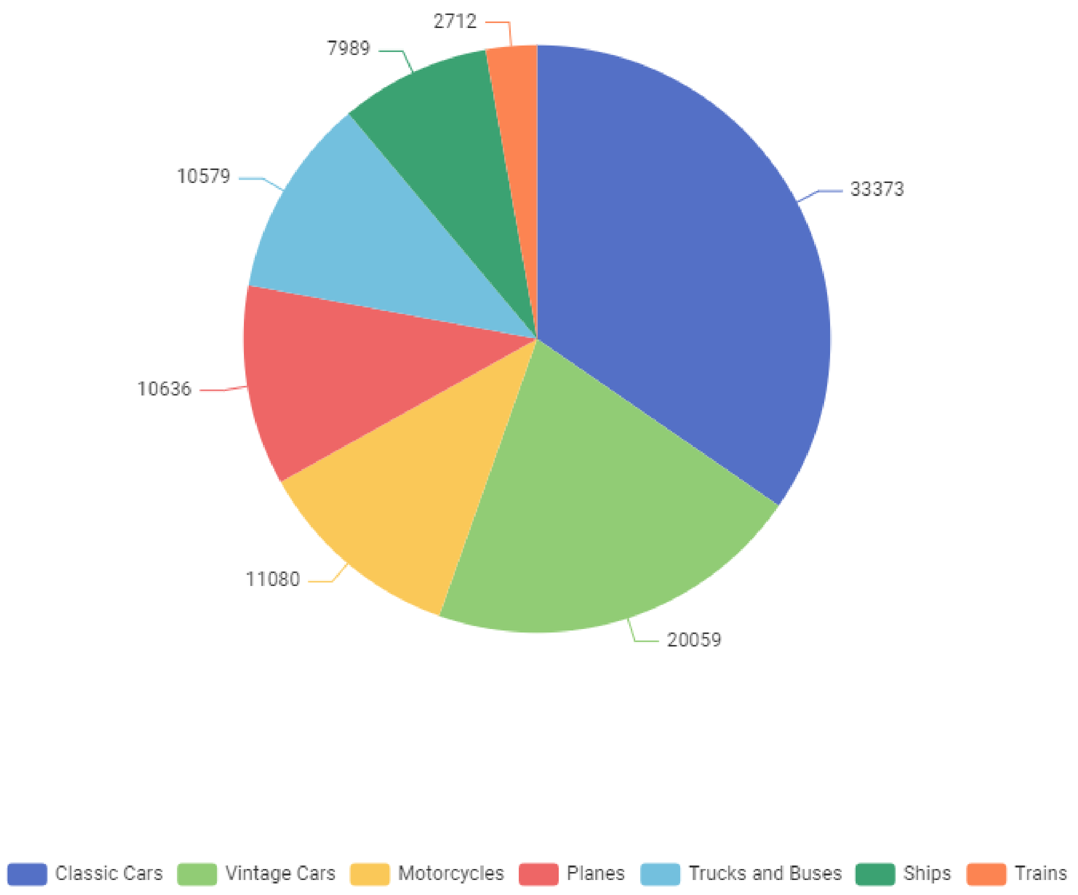


Figure 1: Total Order vs Product Line

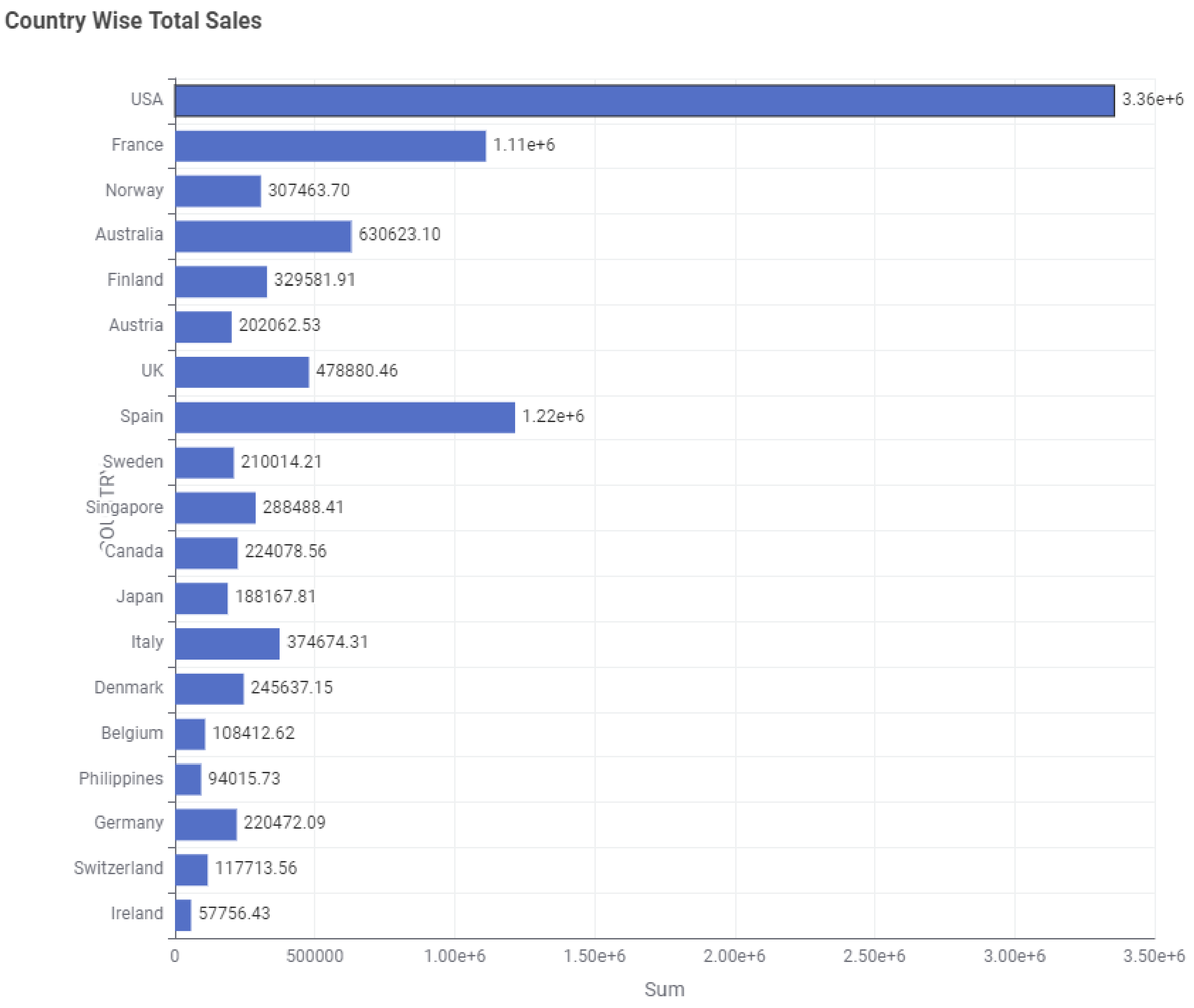


Figure 2:Country VS Total Sales

The USA leads in sales, with approximately $3,360,000. Following the USA are Spain, with $1,220,000 in sales, and France, with $1,110,000. The lowest sales figures were recorded in Ireland ($57,756.43) and the Philippines ($94,015.73).

Orders in the range of 35,000–45,000 accounted for the highest proportion at 23%. In contrast, orders exceeding 68,000 were extremely rare, making up only about 1%. Orders above 55,000 represented approximately 9%, while the remaining order ranges fell between 16% and 17%.

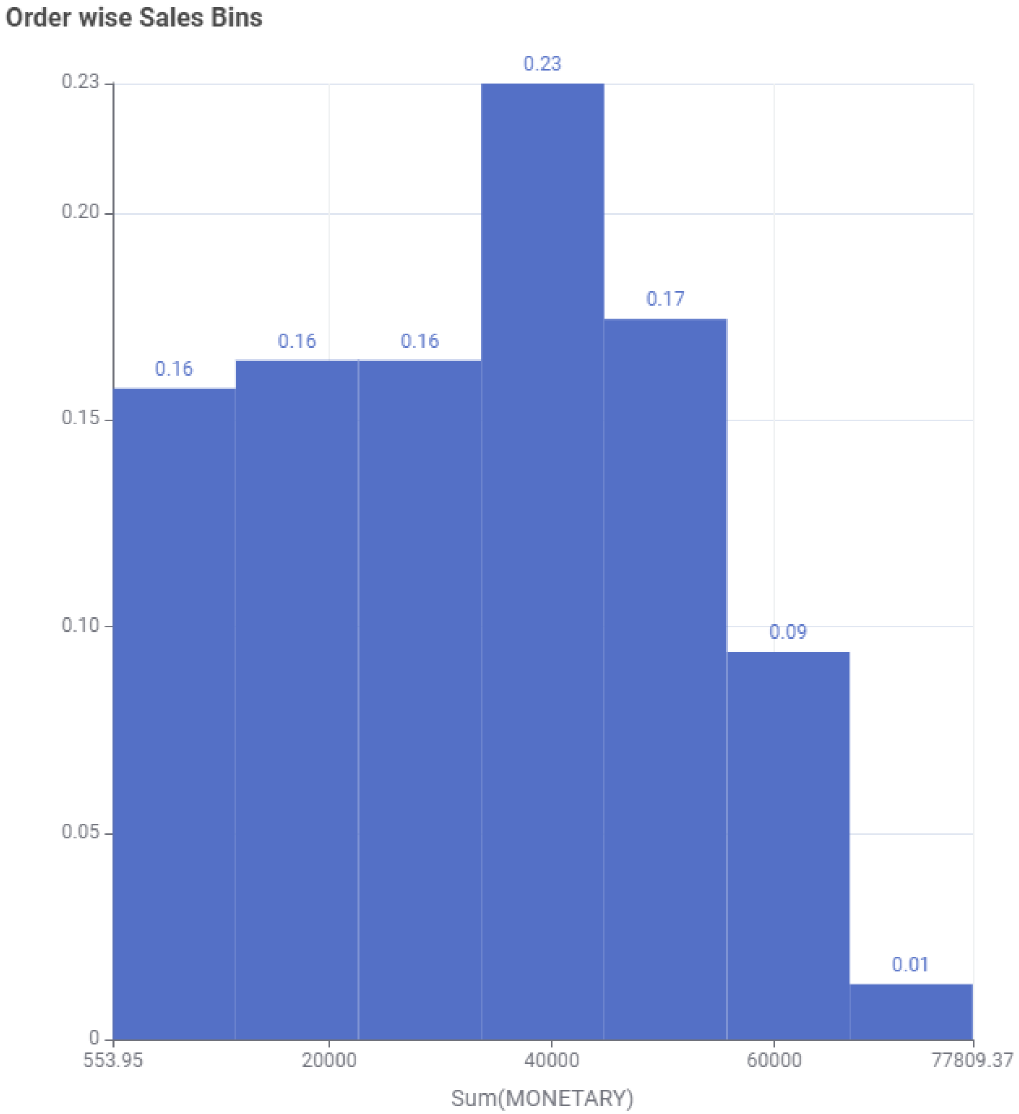


Figure 3: Per Order Sales Bin

Classic cars generated the highest revenue, amounting to $3,842,868.53. The second-highest revenue came from vintage cars, with approximately $1,806,675.67. In contrast, trains recorded the lowest revenue, totaling $226,243.47.

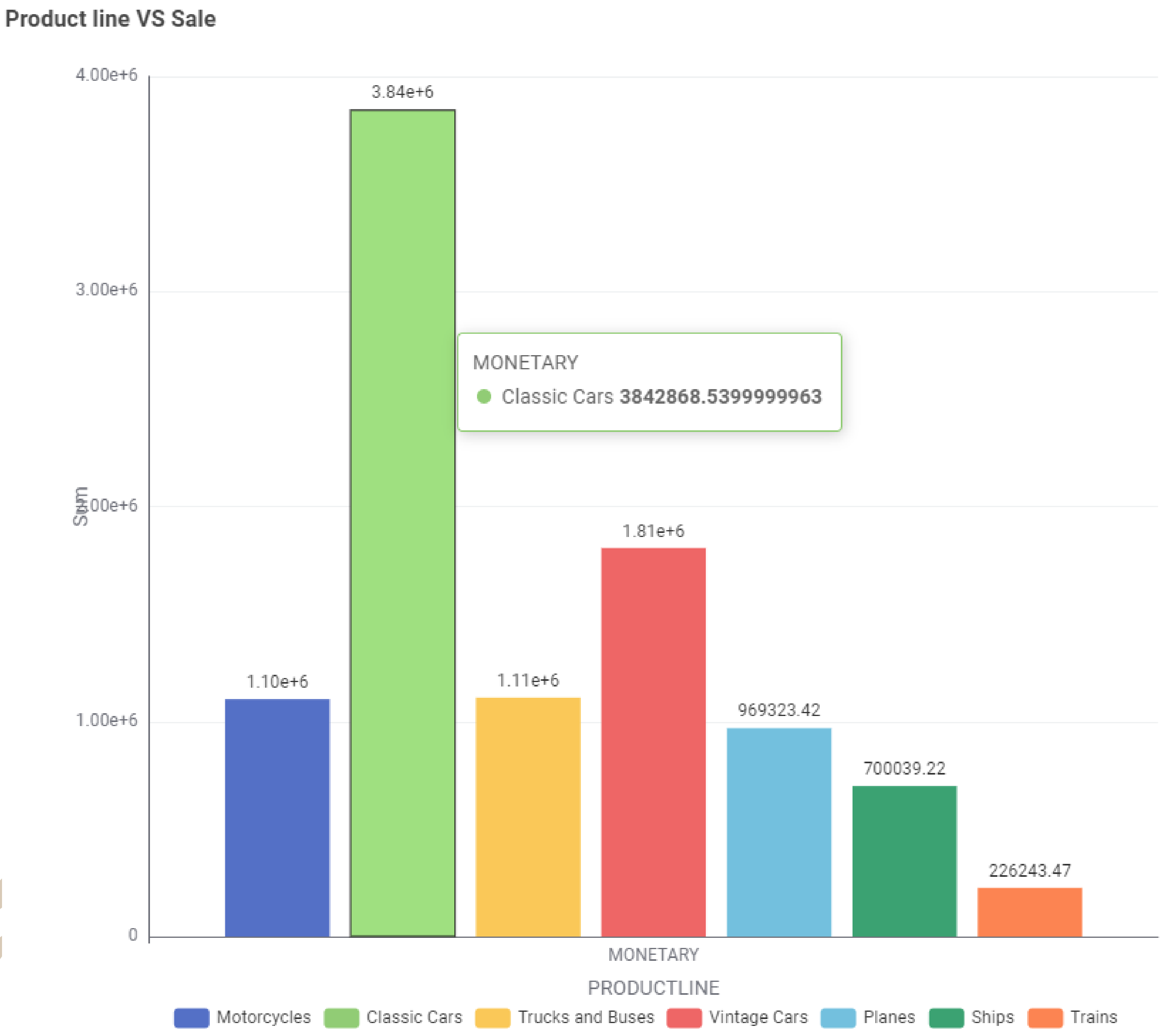


Figure 4: Product line VS Total Sales

The majority of orders were shipped, totaling 2,541, while only a small number were canceled, with just 60 orders.

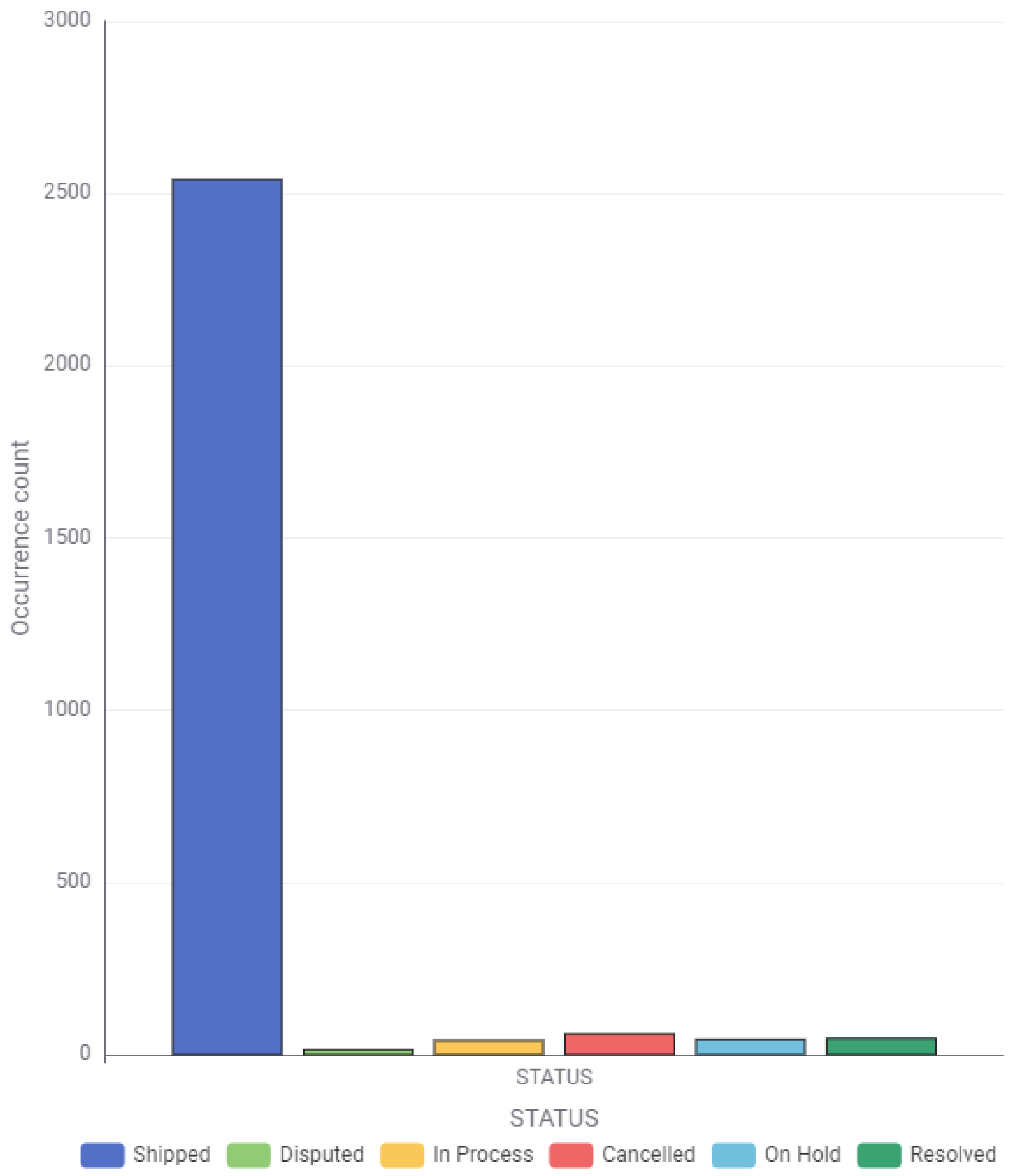


Figure 5:Order status

- Sales totaled $3,300,000 in 2018, increased significantly to approximately $4,700,000 in 2019, and dropped to a low of around $1,700,000 in 2020.

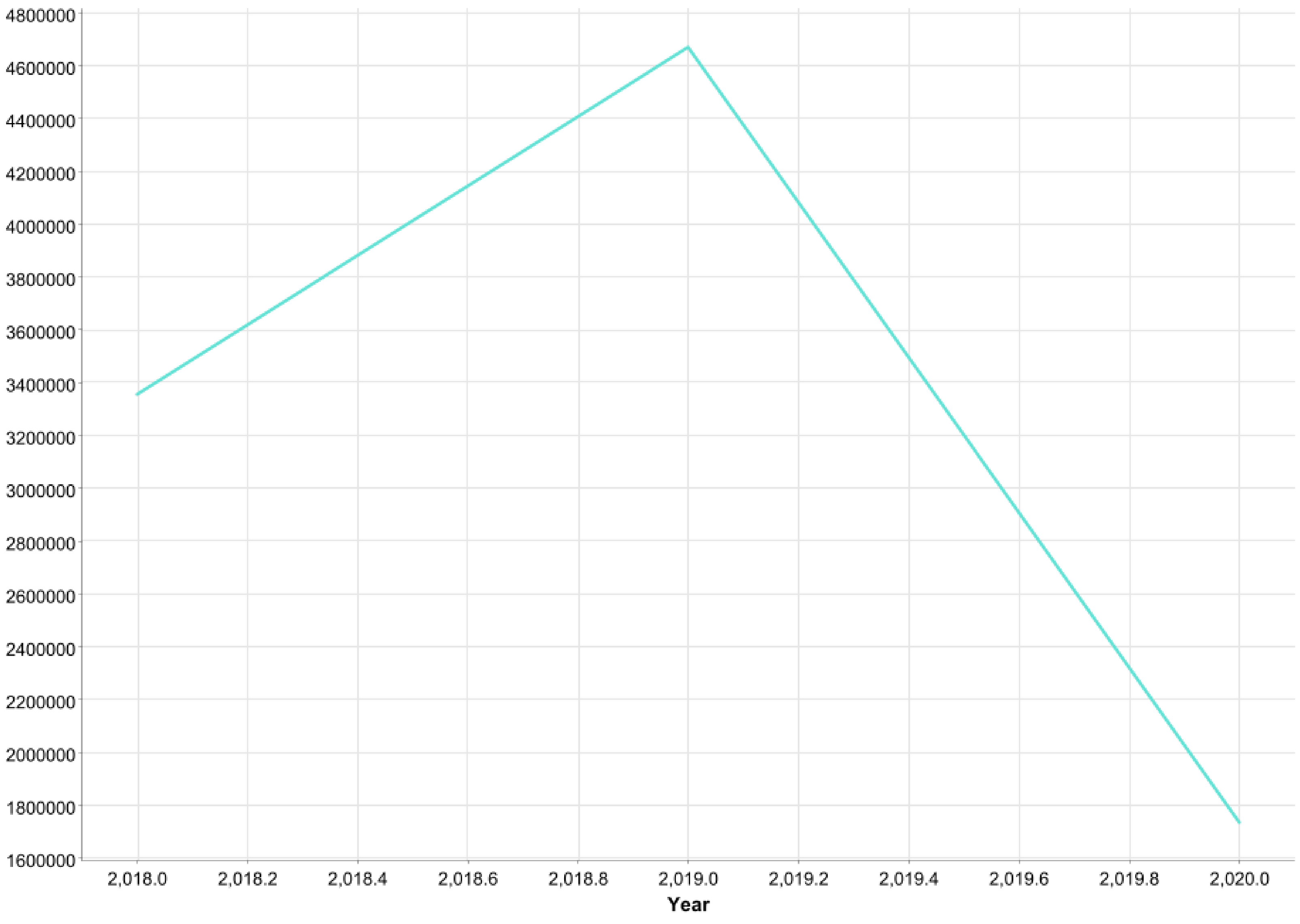


Figure 6: Sum of Sales per Year

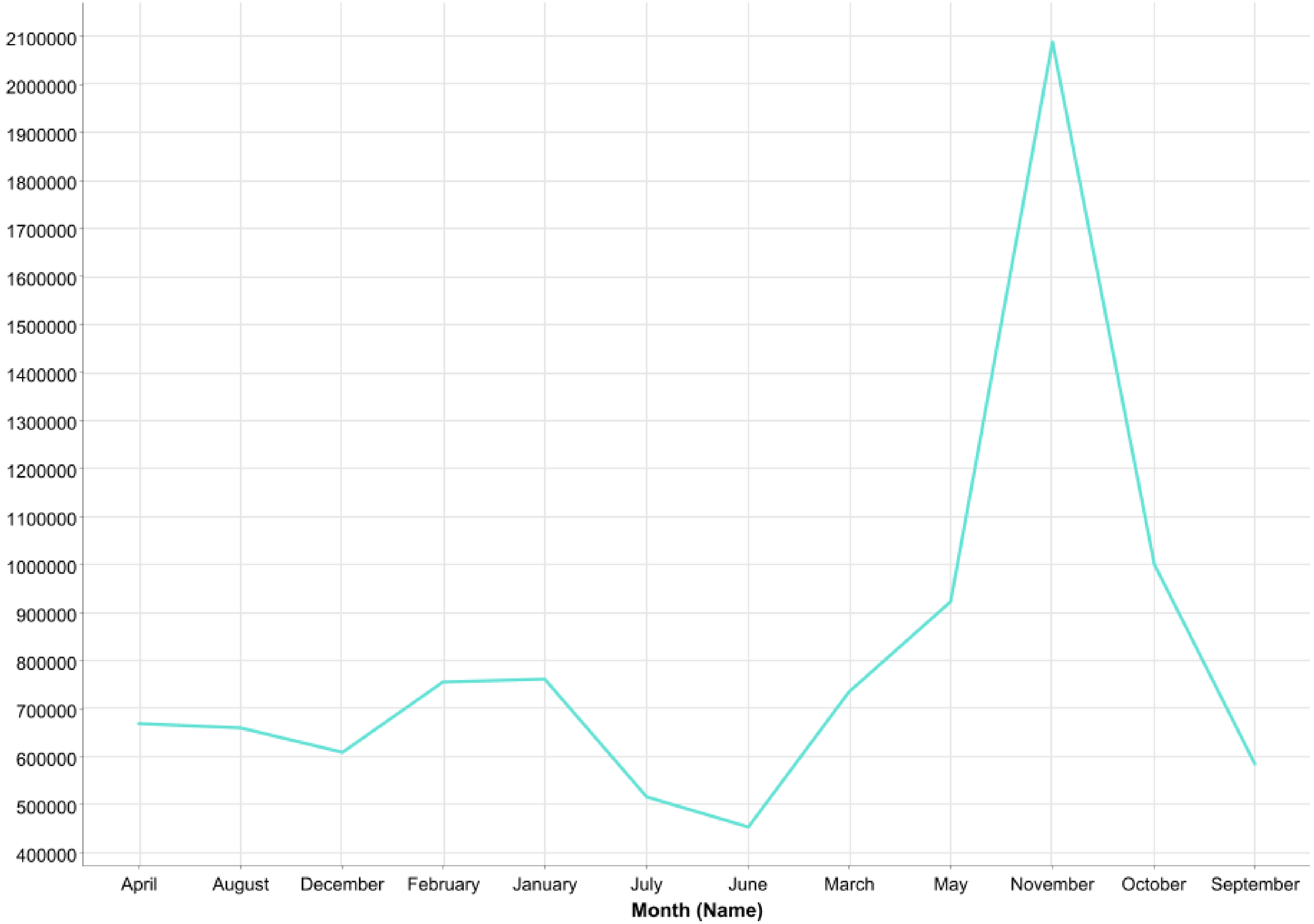


Figure 7: Sum of Sales per Month

- Sales were notably high in November, reaching up to $2,100,000.

- Sales were at their lowest in Quarter 3, totaling $2,250,000, while Quarter 4 recorded the highest sales, reaching $3,700,000.

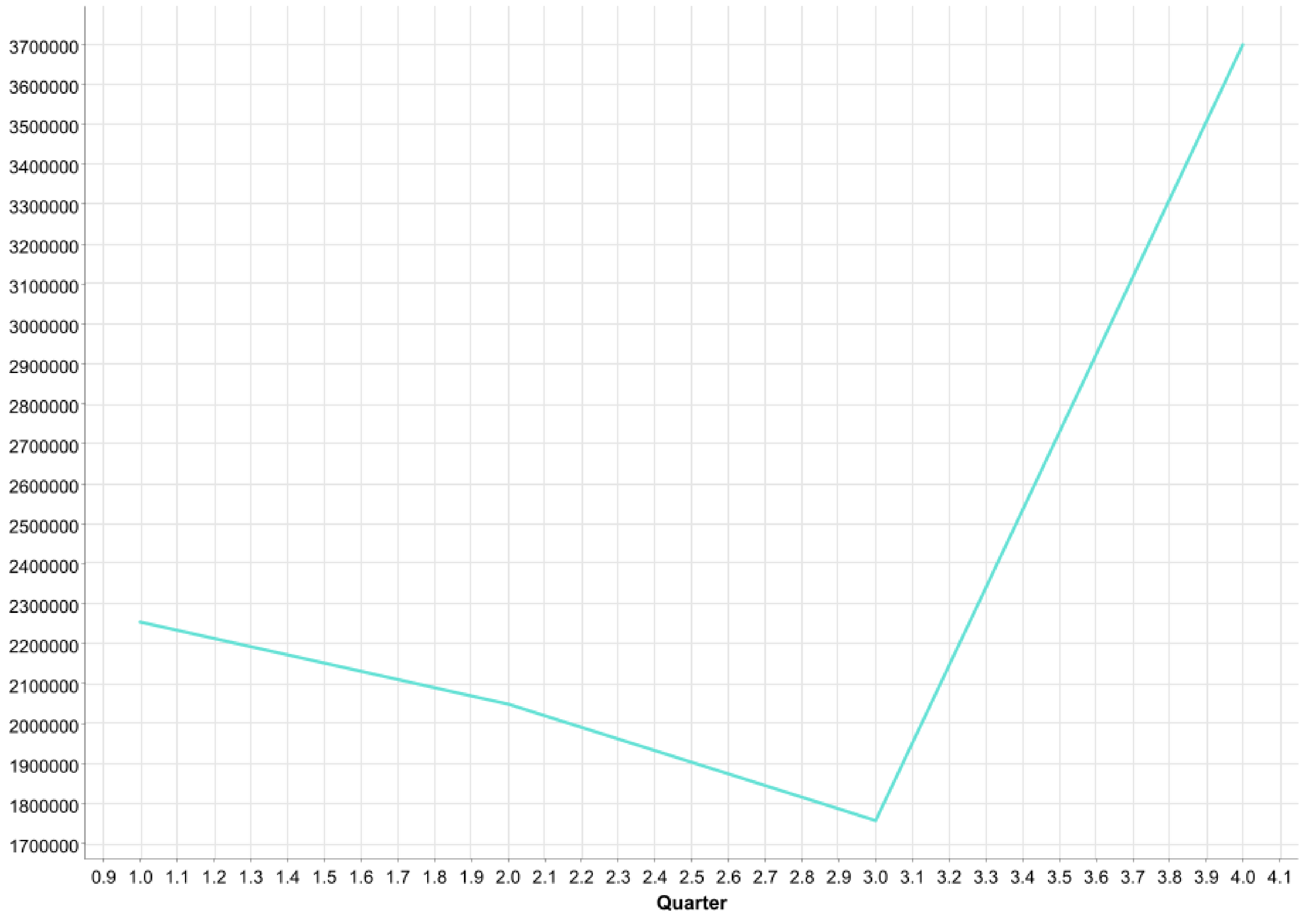


Figure 8; Sum of sales per quarter

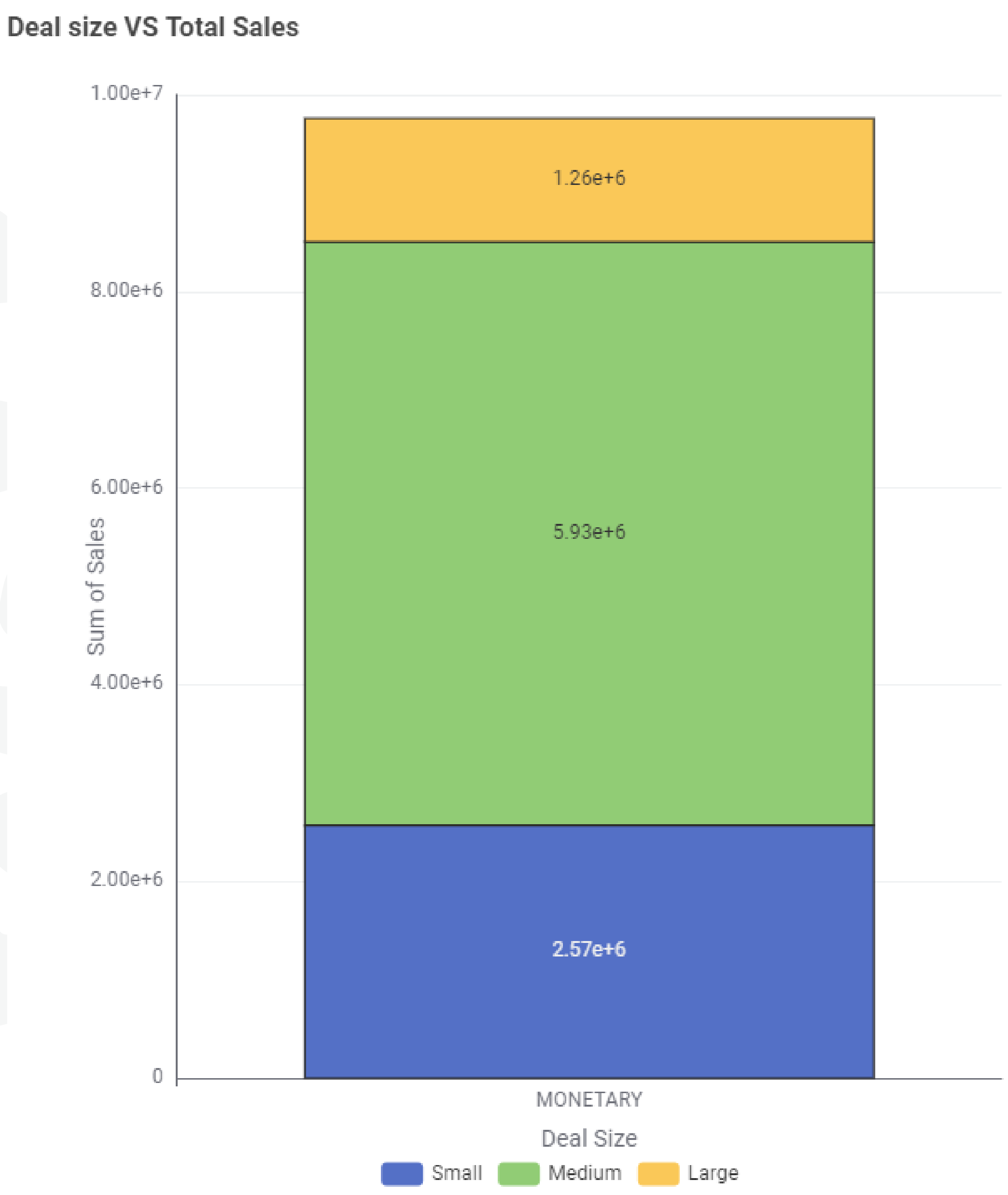


Figure 9: Deal size VS Sales

- Total sales were highest for medium-sized deals, amounting to $5,131,231.47. In comparison, sales were lowest for large-sized deals at $1,258,956.40, while small-sized deals generated a moderate total of $2,570,033.83.

- Dragon Souvenirs, Ltd. is the top customer with a total purchase of $77,809.37. The table highlights the top 10 customers who made the highest purchases, with other notable customers including Toms Spezialitäten, Ltd. ($68,943.40) and Muscle Machine Inc. ($68,462.15).

Order Number	Sum of Sales	Customer name
10165	77809.37	Dragon Souveniers, Ltd.
10310	68943.4	Toms Spezialitten, Ltd
10127	68462.15	Muscle Machine Inc
10287	67281.01	Vida Sport, Ltd
10212	65165.17	Euro Shopping Channel
10204	64316.09	Muscle Machine Inc
10192	63981.45	Online Diecast Creations Co.
10207	63730.78	Diecast Collectables
10142	63088.16	Mini Gifts Distributors Ltd.
10312	63075.08	Mini Gifts Distributors Ltd.

Top 10 customers by total sales amount

Market Basket Analysis (MBA):

Support	Confidence	Lift	Consequent	Implies	Items
0.087248	0.928571429	1.637363	Vintage Cars	<---	[Planes, Ships]
0.114094	1	1.763314	Vintage Cars	<---	[Ships, Classic Cars]
0.124161	0.948717949	1.46486	Classic Cars	<---	[Vintage Cars, Trucks and Buses]
0.130872	0.866666667	1.338169	Classic Cars	<---	[Trains]
0.157718	0.810344828	1.428892	Vintage Cars	<---	[Planes]
0.194631	0.920634921	1.623368	Vintage Cars	<---	[Ships]
0.228188	0.957746479	1.4788	Classic Cars	<---	[Trucks and Buses]

a. High Confidence and Lift for Vintage Cars:

- Customers who purchased both "Planes" and "Ships" were highly likely to also buy "Vintage Cars" (confidence: 92.86%, lift: 1.637).
- "Ships" and "Classic Cars" strongly correlate with "Vintage Cars" purchases, achieving perfect confidence (100%) and the highest lift (1.763).
- "Ships" alone is also a strong predictor for "Vintage Cars" purchases (confidence: 92.06%, lift: 1.623).
- "Planes" alone indicates a lower likelihood but still a notable association with "Vintage Cars" (confidence: 81.03%, lift: 1.429).

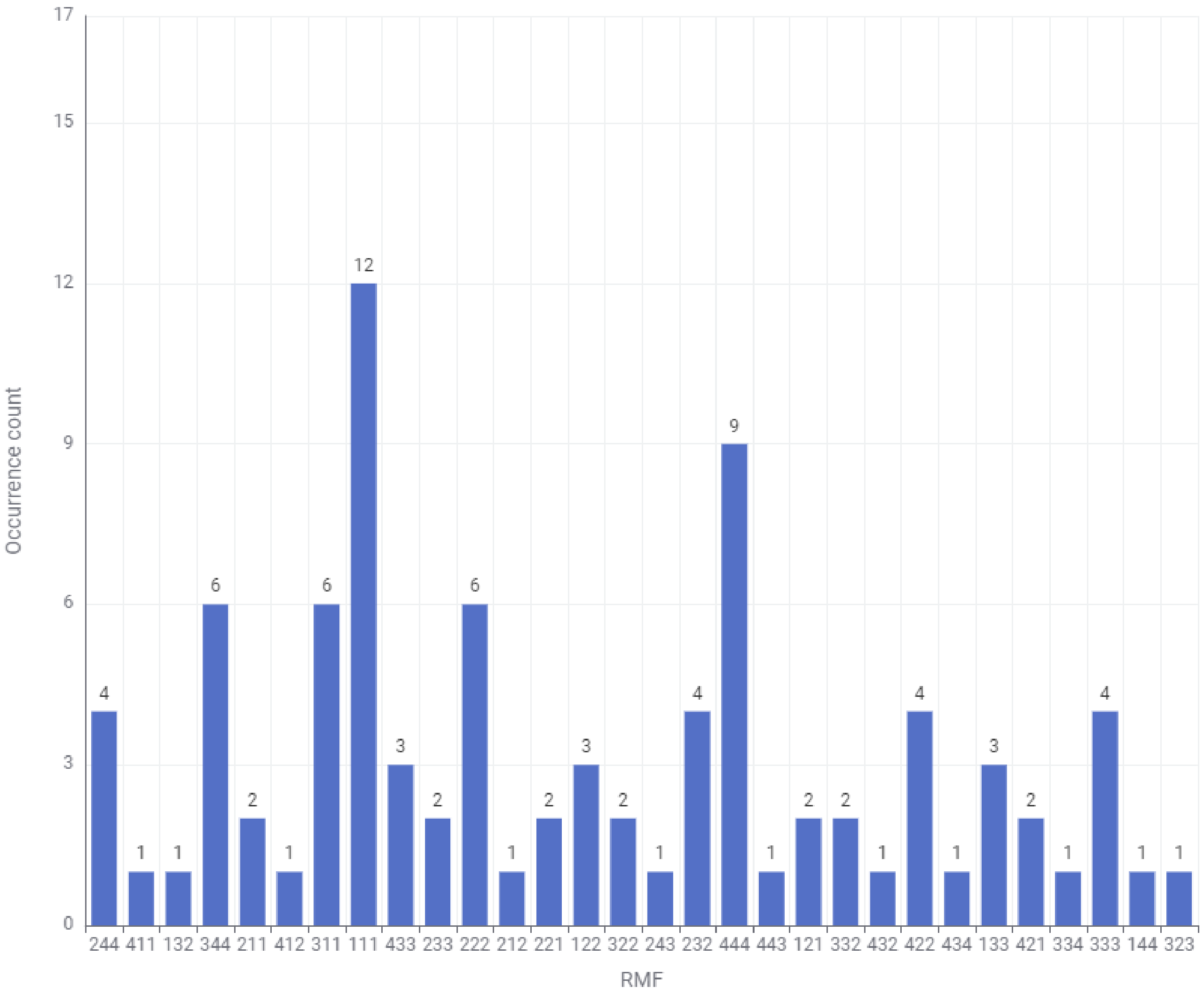
b. Strong Association with Classic Cars:

- Customers who bought "Vintage Cars" alongside "Trucks and Buses" often also purchased "Classic Cars" (confidence: 94.87%, lift: 1.465).
- "Trains" have a moderate association with "Classic Cars" purchases (confidence: 86.67%, lift: 1.338).
- "Trucks and Buses" independently correlate strongly with "Classic Cars" purchases (confidence: 95.77%, lift: 1.479).

c. Dominant Purchase Trends:

- Items like "Planes," "Ships," and "Trucks and Buses" commonly appear in combinations that lead to purchases of "Vintage Cars" or "Classic Cars."
- The highest lift values indicate that "Ships" and "Classic Cars" together are the strongest predictors for "Vintage Cars."

RFM Analysis



Output RFM file

a. **Segment Distribution (Top RMF Scores):**

- The most frequent RFM scores are:
  - **444 (9 customers):** These are the most valuable customers who scored the highest in Recency, Frequency, and Monetary.
  - **344 (6 customers):** Loyal and recent customers with high Frequency and Monetary scores.
  - **311 and 222 (6 customers each):** Customers with moderate Recency, Frequency, and Monetary scores.

b. **Summary of RFM Components:**

- **Recency:** The average score is **2.52**, indicating that most customers have been relatively recent in placing orders.
- **Frequency:** The average score is **2.36**, showing moderate purchase frequency overall.
- **Monetary:** The average score is **2.48**, with a strong spread toward high-value transactions.

Customer Segments:

- **High-Value Customers (444):** Represent the most loyal, frequent, and high-spending customers. Focus retention efforts here.
- **Loyal but Less Frequent (344, 334, 433):** These customers may not spend as much or order frequently but have strong potential for upselling.
- **New or Infrequent Customers (111, 121):** Customers who made one-time or low-value purchases. Engage with them to increase retention and frequency.
- **At-Risk Customers (211, 221):** Moderate spenders with low Recency scores; re-engagement strategies are necessary to bring them back.

Action Items

1. At-Risk Customers:

- **RFM Scores:** Low Recency (1 or 2), moderate Frequency (2 or 3), and moderate to high Monetary (3 or 4).
- **Example RFM Segments: 211, 221, 222, 311.**
- **Reason:** These customers used to spend frequently or make valuable purchases but haven't been active recently. Offering them discounts, loyalty points, or re-engagement campaigns can help win them back.

2. Potential Loyalists:

- **RFM Scores:** Moderate Recency (3), moderate to high Frequency (3 or 4), and moderate Monetary (2 or 3).
- **Example RFM Segments: 333, 344, 334, 433.**
- **Reason:** These customers show good potential for becoming loyal customers if nurtured. Offering them rewards for their loyalty, such as early access to new products or small discounts, can help solidify their loyalty.

3. New Customers:

- **RFM Scores:** High Recency (4), low to moderate Frequency (1 or 2), and low Monetary (1 or 2).
- **Example RFM Segments: 411, 421, 412.**
- **Reason:** These are new or recently acquired customers who haven't yet made many purchases. Giving them welcome offers, first-time discounts, or referral bonuses can encourage more frequent purchases.

4. Low-Value Customers with Potential:

- **RFM Scores:** Low Recency (1 or 2), low Frequency (1 or 2), and low Monetary (1 or 2).
- **Example RFM Segments: 111, 121, 122.**
- **Reason:** These are low-value customers who may have disengaged. Special offers or discounts can encourage them to come back and spend more.

Who Doesn't Need Discounts?

- **Top Customers (444):**
  - These customers are already spending frequently and generating high revenue. Instead of discounts, they should be rewarded with loyalty programs, exclusive benefits, or personalized perks to maintain their engagement without reducing revenue.

Recommendations:

Optimize Marketing Efforts:

- **Focus on Best-Selling Products:** The data shows that Classic Cars and Vintage Cars are the top-selling product lines. Marketing campaigns should focus on promoting these products through targeted ads and bundling offers with other complementary product lines like Planes, Ships, and Trucks, which have a strong association with Classic and Vintage Cars.
- **Leverage High-Selling Countries:** The USA, Spain, and France are top-performing markets. Allocate more marketing resources, such as localized promotions and loyalty programs, to these regions to maintain and grow sales. However, increase efforts in markets with lower sales (e.g., Ireland and the Philippines) by running promotions to encourage new purchases.
- **Capitalize on Seasonal Trends**: The analysis highlights peak sales in Quarter 4 and the month of November. Plan major marketing pushes, such as special offers, discounts, or new product releases, during this period to maximize sales.
- **Tailor Deal Size Marketing:** Medium-sized deals generated the highest revenue. Target customers likely to make medium-sized purchases with tailored offers that highlight the value of bundling products, free shipping, or volume discounts.
- **Focus on Re-engaging At-Risk Customers:** Customers who haven't made purchases recently (identified through RFM analysis) need re-engagement strategies. Offer them time-limited discounts, loyalty rewards, or exclusive deals to encourage them to return.

Recommendations:

Personalized Marketing Strategies:

- **High-Value Customers (444) – Loyalty Program:** These customers are highly loyal and frequent spenders. Rather than offering discounts, focus on providing them exclusive benefits, such as early access to new products, VIP customer services, and personalized thank-you notes or rewards to enhance their experience and maintain retention.
- **Potential Loyalists (333, 344, 334) – Nurture Loyalty:** These customers have moderate frequency but could become loyal with proper engagement. Provide small incentives like early-bird discounts, personalized product recommendations based on past purchases, or loyalty points to encourage repeat purchases.
- **New Customers (411, 421) – Welcome Offers:** For customers who have made recent purchases but show low frequency, offer them first-time buyer discounts, personalized welcome messages, and referral bonuses to encourage them to become repeat buyers.
- **At-Risk Customers (211, 221) – Re-engagement Campaign:** These customers have reduced activity recently. Use personalized email campaigns with attractive discounts or special offers such as “We Miss You” campaigns, targeted specifically at this group, to revive their interest and prompt new purchases.
- **Low-Value Customers (111, 121) – Upselling and Cross-Selling:** These customers are disengaged but could increase their spending. Use targeted product recommendations for upselling and cross-selling based on the market basket analysis. Offering discounts on higher-margin products or bundles can help increase their purchase value.

Cont...

Cross-Selling and Product Bundling:

- **High Confidence and Lift Relationships:**
  - Customers who purchase "Planes" and "Ships" are very likely to buy "Vintage Cars" (92.86% confidence, 1.637 lift). To take advantage of this, create bundle offers or cross-sell promotions that feature "Vintage Cars" alongside "Planes" and "Ships."
  - "Ships" and "Classic Cars" also show a strong correlation with "Vintage Cars" (100% confidence, 1.763 lift). Offer discounts on "Vintage Cars" when a customer buys "Ships" or "Classic Cars" together, creating a tailored package.
- **Recommendation**: For online marketing and email campaigns, recommend these associated products based on a customer's current cart or purchase history. For example, if a customer adds "Planes" or "Ships" to their cart, the system should recommend "Vintage Cars" with a personalized discount or free shipping on the second item.

Targeted Promotions for Complementary Products:

- **Trucks and Buses + Vintage Cars → Classic Cars (94.87% confidence, 1.465 lift):**
  - Promote targeted ads or personalized email campaigns that encourage customers who have purchased Trucks and Buses along with Vintage Cars to buy Classic Cars.
  - **Example Campaign**: "Complete your collection: Add a Classic Car to your Trucks and Vintage Cars purchase and get 10% off!"
- **Planes + Vintage Cars → Classic Cars (86.67% confidence, 1.338 lift):**
  - When customers buy Planes or Vintage Cars, suggest Classic Cars as a complementary purchase. This can be part of a holiday promotion or a limited-time offer campaign to drive additional sales.

Cont...

Focus on High-Value Item Bundles:

- **Ships and Classic Cars (highest lift of 1.763):**
    - Since this combination has the strongest relationship in the MBA, offering bundle promotions or discounts that encourage customers to buy these items together can increase average transaction values. Use "Buy X, Get Y" offers or discounted packages that bundle these items.

Personalized Upselling and Recommendations:

- Based on customers' purchase history, send personalized product recommendations that highlight complementary products frequently bought together. For example, if a customer buys "Planes," suggest adding "Vintage Cars" to their order with a small incentive (e.g., 5% discount if they complete their purchase within 24 hours).

Sales Floor and E-Commerce Website Optimization:

- **In-store Promotions and Website Layouts:** Place products with high association (e.g., Planes and Ships, Vintage Cars and Classic Cars) near each other in physical stores or on related pages online. This increases visibility and the likelihood of customers buying both items.

- **Example Layout Optimization**: When a customer browses for "Ships," feature "Vintage Cars" prominently on the same page or provide a pop-up offering a bundle deal. Similarly, when customers purchase "Trucks and Buses," show related "Classic Cars" as part of a recommended section.

Special Offers on Items with Lower Lift but Strong Potential:

- **Trains and Classic Cars (86.67% confidence, 1.338 lift):**
    - Though this association has lower lift compared to others, there's still potential for promotion. Offer discounts or loyalty points when customers purchase these items together to increase the likelihood of future purchases in this segment.

Part B

- **Business Context:**

• In the highly competitive grocery retail industry, understanding customer buying patterns is crucial for enhancing sales, increasing customer satisfaction, and improving profitability. By identifying frequently purchased item combinations, grocery stores can craft effective marketing strategies, optimize inventory management, and tailor promotions to meet customer needs. Leveraging Point of Sale (POS) data can unlock valuable insights that drive customer-centric offerings, such as combo packs, discounts, and targeted promotions, which can increase basket size and improve customer retention. This analysis aligns with business goals by maximizing revenue, reducing operational costs, and boosting customer loyalty.

- **Objective:**

• As a business analyst, the goal is to analyze the POS transactional data to identify frequently purchased item combinations. Using association rule mining or similar techniques, the aim is to uncover patterns that will help the store create targeted combo offers and discounts, ultimately driving revenue growth by increasing customer purchases and average basket size.

- **Data Description:**

- **Date:** The date when the transaction took place.
- **Order_id:** A unique identifier for each customer order.
- **Product:** The individual item purchased in the transaction.

Market Basket Analysis (MBA):

Support	Confidence	Lift	Consequent	Implies	Items
0.055312	0.649484536	1.79119343	paper towels	<---	[eggs, ice cream, pasta]
0.055312	0.642857143	1.73100304	pasta	<---	[paper towels, eggs, ice cream]
0.053556	0.642105263	1.650920756	dinner rolls	<---	[spaghetti sauce, poultry, laundry detergent]
0.0518	0.641304348	1.648861517	dinner rolls	<---	[spaghetti sauce, poultry, ice cream]
0.0518	0.686046512	1.627931202	poultry	<---	[dinner rolls, spaghetti sauce, ice cream]
0.0518	0.634408602	1.627458103	eggs	<---	[paper towels, dinner rolls, pasta]
0.0518	0.602040816	1.621098085	pasta	<---	[paper towels, eggs, dinner rolls]
0.055312	0.63	1.616148649	eggs	<---	[paper towels, ice cream, pasta]
0.0518	0.627659574	1.613779357	dinner rolls	<---	[spaghetti sauce, poultry, juice]
0.0518	0.627659574	1.610144719	eggs	<---	[dinner rolls, poultry, soda]
0.055312	0.623762376	1.564901644	ice cream	<---	[paper towels, eggs, pasta]
0.053556	0.655913978	1.556429211	poultry	<---	[dinner rolls, spaghetti sauce, laundry detergent]
0.0518	0.602040816	1.428592687	poultry	<---	[dinner rolls, spaghetti sauce, juice]

Support: This indicates the proportion of transactions that contain the items mentioned in the rule. For example, a support value of 0.0553 for the rule [eggs, ice cream, pasta] → paper towels means that 5.53% of all transactions in the dataset contain the combination of "eggs, ice cream, and pasta" and also include "paper towels."

Confidence: Confidence measures how often the rule has been found to be true. For instance, the confidence of 0.649 for the rule [eggs, ice cream, pasta] → paper towels suggests that 64.9% of the transactions that include "eggs, ice cream, and pasta" also include "paper towels." High confidence implies that the presence of the items on the left-hand side (LHS) of the rule is a strong indicator of the presence of the item on the right-hand side (RHS).

Lift: Lift measures the strength of the association between the LHS and RHS, comparing the observed co-occurrence with how frequently the items would co-occur by chance. A lift value greater than 1 indicates a positive association (the items are more likely to be bought together than by random chance). For instance, the lift of 1.791 for the rule [eggs, ice cream, pasta] → paper towels indicates that these items are 1.79 times more likely to be bought together than would be expected if they were independent of each other.

Interpretation of Association Rules

- **Rule: [eggs, ice cream, pasta] -> paper towels**
  - **Support**: 5.53% of transactions contain all these items together.
  - **Confidence**: In 64.9% of the transactions where "eggs, ice cream, and pasta" are bought, "paper towels" are also purchased.
  - **Lift**: These items are 1.79 times more likely to be bought together compared to being bought independently.
- **Rule: [spaghetti sauce, poultry, laundry detergent] -> dinner rolls**
  - **Support**: 5.36% of transactions include this combination.
  - **Confidence**: In 64.2% of cases where these items are bought, dinner rolls are also purchased.
  - **Lift**: 1.65, indicating a strong association.
- **Rule: [dinner rolls, spaghetti sauce, ice cream] -> poultry**
  - **Support**: 5.18% of transactions.
  - **Confidence**: 68.6%, indicating that poultry is commonly purchased with these items.
  - **Lift**: 1.63, showing a meaningful relationship between these items.
- **Rule: [paper towels, eggs, pasta] -> ice cream**
  - **Support**: 5.53% of transactions.
  - **Confidence**: 62.4%, meaning ice cream is often bought with this combination.
  - **Lift**: 1.56, a positive association indicating frequent co-purchase.

Recommendations:

1. **Create Combo Offers for Commonly Bought Items**: Based on the association rules, the store should create combo offers for frequently purchased items such as "eggs, ice cream, and pasta" with "paper towels." By offering discounts or incentives on these combinations, the store can encourage larger basket sizes.

2. **Targeted Discounts for Complementary Items**: Rules like [spaghetti sauce, poultry, laundry detergent] → dinner rolls show that customers tend to buy these items together. Offering discounts or buy-one-get-one-free deals on "dinner rolls" when a customer buys "spaghetti sauce and poultry" can encourage additional purchases.

3. **Cross-Promote Items with High Lift Values**: Items with high lift values, such as "eggs, ice cream, and pasta" with "paper towels" (lift of 1.79), should be cross-promoted in-store and online. This could involve placing these items near each other in the store or offering bundled discounts online.

4. **Segment Customers for Personalization**: Use the association rules to segment customers based on their purchase history. For example, if a customer frequently buys "poultry and spaghetti sauce," offer them personalized discounts on "dinner rolls" or related items.

Recommendations:

Example Discounts

1. **"Breakfast Combo" Discount**:
   - *Buy any combination of eggs, ice cream, and pasta, and get 10% off on paper towels.*
   - Encourages larger purchases while providing value to the customer.
2. **"Dinner Essentials Offer"**:
   - *Buy spaghetti sauce and poultry, and get 50% off on dinner rolls.*
   - This aligns with the rule [spaghetti sauce, poultry, ice cream] → dinner rolls.
3. **"Grocery Basics Pack"**:
   - *Purchase paper towels, eggs, and pasta together, and get a discount on ice cream.*
   - This promotes items commonly bought together, driving higher basket value.
4. **"Laundry and Poultry Special"**:
   - *Buy laundry detergent and get a discount on poultry or dinner rolls.*
   - This leverages the association between these items found in the data.

THANK YOU

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https://arisa365.github.io/